

Exercise - Week 8: MATH1061/7861

Duration: 30 minutes

7 Questions - 10 marks

Topics: Properties of Sets; Functions Defined on General Sets;
One-to-One, Onto and Inverse Functions; Function Composition

Student Name:

Student number: |_|_|_|_|_|_|_|_|_|_|

Your student number should be 8 digits and **not** start with an s. e.g. 41234567

Let $f : \mathbb{Z} \rightarrow \mathbb{Z}$ be defined by $f(x) = 2x^2 - 1$.

Let $g : \mathbb{Z} \rightarrow \mathbb{Z}$ be defined by $g(y) = y + 2$.

(1) (1 mark) What is the image of $\mathbb{Z}_{\geq 2}$ under g ?

(2) (1 mark) Determine $f^{-1}(7)$.

(3) (2 marks) Is f one to one? Justify your answer.

(4) (2 marks) Is f onto? Justify your answer.

(5) (**1 mark**) Determine the composition $f \circ g$.

(6) (**1 mark**) Determine the composition $g \circ f$.

(7) (**2 marks**) Is $g \circ g$ is bijective? Justify your answer.